Capstone 12: Submission Pack

EV Chips (SiC/GaN): verification report, hallucination log, prompt library, before/after analysis and demo plan

AI Skills Course (Topics 1-9) · Electronics and Communication Engineering · Prepared by Abinaya R · 25 October 2026

1. Submission checklist

Required item	Where it is
Research questions	Section 2 of this pack and Section 2.1 of the report
Source log	Capstone12_Source_Log_v2.xlsx (39 sources used, 9 not used; automated status counts) and Section 6
Research report (6-8 pages, citations)	Capstone_12_EV_Chips_Research_Report.pdf (8 pages including references)
Charts (3)	Inside the report: Figures 1-3
Presentation (5-7 slides)	Capstone_12_EV_Chips_Executive_Summary_v2.pptx (6 slides)
Verification report	Section 4
Hallucination and unsupported-claim log	Section 5
Prompt library	Section 7
Before/after analysis	Section 8
Copyright, citation practice, limitations	Section 9
Final demo	Section 10

2. Research questions

- **RQ1.** Which chips are used in an EV (power, battery management, motor control), and what do SiC and GaN offer over silicon in efficiency and cost?
- **RQ2.** Who controls the EV chip supply chain (Infineon, STMicroelectronics, onsemi and others), and what is India's position (ISM, Tata Electronics, CG Power)?
- **RQ3.** What are the main trends and risks (shortage, cost, import dependency)?

3. How the common standard is met

Requirement	Evidence in this project
6 or more AI skills (Topics 1-9)	Prompt engineering; AI-assisted research; source verification; multi-tool comparison; report writing; data visualisation (charts); presentation design; workflow automation (source log); responsible-AI analysis
4 or more AI tools	Perplexity, Claude, ChatGPT and Gemini for research; Claude also for drafting, charts and slides; a spreadsheet for the automated source log
Research component	The same three questions run in four tools; 39 sources used and 9 rejected
Content generation	8-page report, executive summary, slide text
Multimodal component	Three charts and a 6-slide deck
Automation / workflow	Source log with status drop-down, automatic date stamp, status counts and an open-action flag
Prompting, verification, error analysis	Sections 4, 5 and 7
Engineering verification rule	Every AI-supplied value was checked against a page or recomputed by hand (Sections 4.3 and 4.4); values that could not be checked were dropped
Responsible AI; before-vs-after; live demonstration	Sections 9, 8 and 10

4. Verification report

4.1 Method

Each AI answer was split into single claims. Each claim got a verdict: **Verified** (a source supports it), **Partly verified**, **Conflict** (sources or a hand calculation disagree), **Unsupported** (no source found) or **Not checked**. A claim is used in the report only if it is Verified or Partly verified with its limits stated. After drafting, every cited source was opened again on 24-25 October 2026 and compared with the exact claim it supports (Section 4.3).

4.2 Results by tool

Perplexity (20 claims checked).

Question	Verified	Partly	Conflict	Unsupported	Not checked
Q1 chips and SiC/GaN	4	0	0	1	1
Q2 supply chain and India	3	1	0	0	3
Q3 trends and risks	2	2	1	1	1
Total	**9**	**3**	**1**	**2**	**5**

Pattern: well-structured tables, but citations were site names without URLs, one number had no source (12-18 months), one row contradicted another (bottleneck versus oversupply), and the India part of Q2 and the SiC advantage part of Q1 were not answered.

Claude. Every figure came with a numbered source and URL. Conflicts were flagged (ISM project count; ₹7,600 crore versus ₹7,584 crore). Weaknesses: EDA/IP, foundries and ADAS chips not researched, several vendor or trade sources, and the August 2025 construction status was out of date by October 2026.

ChatGPT (short Tanglish guide). Mostly accurate and cautious, with five official or company links.

Claim	Verdict	Note
EV chip groups: power devices, MCU and gate drivers, BMS chips, vehicle electronics	Verified	[2][3][4]
SiC suits high-voltage paths; GaN suits compact fast converters; silicon stays cost-competitive	Verified	[6][9][10]
GaN traction use is less mature than SiC	Partly verified	DOE draft names cost as a GaN limit [10]
Tata-PSMC Dholera fab and Tata Assam assembly-and-test	Verified	[14]
CG Power + Renesas + Stars OSAT at Sanand	Verified	[14]
India imports about 90-95% of semiconductor demand	Verified	NITI Aayog [17]
Project approval is not high-volume output	Verified in spirit	Matches the status evidence [28][32]

Gemini (technical deep dive, no sources). Most numbers had no citation.

Claim	Verdict	Note
EVs need 1,500-3,000+ chips versus 300-500 in ICE cars	Unsupported	No source found
Bandgap 1.1 / 3.26 / 3.4 eV; thermal conductivity 1.5 / 4.9 / 1.3 W/cm-K	Verified; GaN thermal value differs	Bandgap and Si, SiC conductivity match [39][40]; GaN is 1.3 on [40] but 2.0 on [39] (C7)
SiC gives 50-70% lower loss, GaN 60-80%	Unsupported	Paper reports over 40% conduction and over 60% switching loss reduction for SiC [1]; no GaN figure found
SiC adds 5-10% range	Unsupported	No source
Chip cost 2.5-3× (SiC) and 2-2.5× (GaN) of silicon	Unsupported	Sources say only "higher"
Battery pack 5-8% smaller; pack cost $100-$130/kWh	Unsupported	No source

SiC for 400/800 V inverters; GaN for DC-DC	Partly verified	[6]: 650 V GaN suits 400 V; 800 V favours 1,200 V SiC
800 V enables 10-80% charge in 15 minutes	Unsupported	Only the 800 V voltage fit is sourced [6]
India imports over 90% of automotive chips	Partly verified	NITI figure is for all semiconductor demand [17]
ISM incentive ₹76,000 crore	Verified	[29]
Tata fab at Dholera with PSMC; OSAT at Jagiroad, Assam	Verified	[14]; Jagiroad is in Morigaon district
CG Power-Renesas "JV" at Sanand	Partly verified	Third partner Stars Microelectronics omitted [14]
Kaynes SemiCon and Micron set up memory and power packaging	Partly verified	Kaynes OSAT ₹3,307 crore [16]; Kaynes-Micron link and "memory and power" wording not verified
Infineon is the world leader in automotive power semiconductors	Partly verified	ST leads SiC devices [38]; Infineon's overall automotive ranking not checked
ST is a major SiC supplier to Tesla	Not checked	ST does lead SiC devices [38]; the Tesla link has no checked source
Wolfspeed and ROHM are pure-play SiC leaders	Partly verified	Wolfspeed leads substrates [11]; ROHM is among the top five in devices [38]
$1,200+ chip value per EV by 2030 versus about $500 for ICE	Unsupported	No source
200 mm SiC wafers give 1.4× more dies and about 30% lower cost	**Conflict**	Area ratio is (200 ÷ 150)² = 1.78; the 30% cost drop has no source
SiC substrate shortage	**Conflict**	TrendForce reports 2024 oversupply and a 9% revenue fall [11]
Silver sintering, zonal architecture, OEM vertical integration	Not checked	Plausible trends; no source in this project

4.3 Final re-check of every cited source (24-25 October 2026)

The first 34 used sources were reopened and compared with the exact claim each supports. Six new sources were added during the re-check, giving 40 numbered references; [19] was withdrawn.

Result	Count	Sources and action
Passed outright	23	[1][2][3][4][6][7][8][9][10][11][13][14][15][17][18][20][21][24][26][27][29][30][34]; small wording notes applied in the report (for example "simulated", "up to ASIL C", "temporarily suspended")
Outdated	1	[31] ISM 2.0 was "pending Cabinet approval"; the Cabinet approved it on 15 July 2026 [35]. Text updated
Partly supported	5	[12] no onsemi 23.6% or Infineon 16.5% on the page (shares removed; ST 32.6% and top-five 91.9% now cite [38]); [22] 2021 page, no export-control claim (moved to [37]); [25] does not name carmakers (reworded); [32] no ground-breaking date (now "early November 2025"); [33] approval by the Falta SEZ Unit Approval Committee, not "the Odisha government" (reworded)
Not supported	1	[19] no import figures on the page. Withdrawn; FY25 import claim removed
Could not be opened	4	[28] was a "test" subdomain; replaced by the official PIB release on pmindia.gov.in. [5], [16], [23] stay as supporting detail and are flagged in the source log and in the report limitations

4.4 Hand calculations

Check	Working	Result
SiC substrate top four, 2024	33.7 + 17.3 + 17.1 + 13.9	82.0%; matches TrendForce [11]
Chinese substrate vendors	17.3 + 17.1	34.4%

Device share outside the top five, 2023	100 − 91.9	8.1% (derived) [38]
200 mm versus 150 mm wafer area	(200 ÷ 150)²	1.78×; Gemini's 1.4× is not supported
SiCSem annual output	4.8 million diodes + 91.2 million MOSFETs	96 million; matches the 96 million packaged units [34][36]
CG Power rounding	(7,600 − 7,584) ÷ 7,584	0.2%; same project
ISM 2.0 versus ISM 1.0 outlay	127,500 ÷ 76,000 − 1	About 67.8% [35][29]
SiC fab versus silicon fab investment	3,406 ÷ 91,000	3.7%, or 1 : 26.7 (Figure 3)
Bandgap ratio, SiC to Si	3.26 ÷ 1.12	2.9× [39]
Thermal conductivity ratio, SiC to Si	4.9 ÷ 1.5	3.3× [39]

4.5 Conflicts between sources

ID	Topic	What sources say	Resolution
C1	SiC efficiency gain over IGBT	Over 1% [1]; up to 3% [8]; 3-5% (vendor blog)	Quote [1] and [8] with conditions; reject the blog
C2	SiC substrate: shortage or oversupply	Shortage (Perplexity, Gemini); oversupply [11]	Use TrendForce; automotive-grade parts may still be tight [21]
C3	Number of approved ISM projects	10 (Aug 2025); six (blog); 12 (May-Jul 2026); 13 across ISM and SPECS	Resolved: 12 approved units, over ₹1.64 lakh crore, per the 15 July 2026 release [35]
C4	SiC substrate shares	TrendForce 2024 versus an older 2023 article quoting Wolfspeed near 60%	Use TrendForce 2024 and state the year
C5	SiCSem investment	About ₹2,000 crore (US$220 million) [32] versus ₹3,406 crore [33][34]	Unresolved; both quoted with sources
C6	CG Power investment	₹7,600 crore [14] versus ₹7,584 crore [16]	Rounding of one project
C7	GaN thermal conductivity	1.3 W/cm-K [40] versus 2.0 W/cm-K [39]	Unresolved; range quoted
C8	onsemi and Infineon device shares	23.6% and 16.5% in AI output	Not on the TrendForce pages [38][12]; removed

5. Hallucination and unsupported-claim log

ID	Tool	Claim or issue	Type	Status
H1	Perplexity	Qualification takes 12-18 months	Invented number	Unsupported; rejected
H2	Perplexity	Substrate bottleneck and oversupply in one table	Internal contradiction, outdated	Conflict; resolved with [11] [21]
H3	Perplexity	Citations such as "mitsui+1" with no URL	Uncheckable citation	Unverified
H4	Perplexity	BMS uses CAN/LIN and prevents thermal runaway	Not found in sources	Unverified; not used
H5	Perplexity	India part of Q2 answered only as "emerging"	Incomplete answer	Fixed in report
H6	Perplexity	SiC/GaN advantage part of Q1 skipped	Incomplete answer	Fixed in report
H7	Gemini	200 mm wafers give 1.4× more dies	Wrong geometry (true area ratio 1.78)	Conflict; not used
H8	Gemini	Range +5-10%, battery −5-8%, $100-$130/kWh	Invented or unsourced figures	Unsupported; not used
H9	Gemini	Chip cost ratios 2.5-3× and 2-2.5×; loss cuts 50-70% and 60-80%	Unsourced specifics	Unsupported; not used
H10	Gemini	$1,200+ per EV by 2030 versus about $500	Unsourced projection	Unsupported; not used

H11	Gemini	Kaynes and Micron presented as one packaging effort	Conflation of two companies	Partly verified
H12	Gemini	No sources at all	Missing citations	Unverifiable
H13	Claude	Four projects under construction (Aug 2025)	Outdated by Oct 2026	Updated: two projects shipping by May 2026 [28]
H14	Claude	Argus page used without a confirmed date	Uncertain date	Resolved: page is from 2021; used only for system-level savings
H15	All tools	Market shares are for 2023-2024	Stale-data risk	Flagged with year
H16	Claude (draft)	onsemi 23.6% and Infineon 16.5% attributed to the EE Times Asia page	Citation does not support claim	Removed; ST 32.6% and top five 91.9% cite [38]
H17	Claude (draft)	China export controls attributed to an Argus 2021 page	Citation does not support claim	Moved to [37]
H18	Claude (draft)	News agency cited through a "test" subdomain	Unsafe citation	Replaced by the official PIB release [28]
H19	Claude (draft)	ISM 2.0 "pending Cabinet approval"	Outdated status	Updated: approved 15 July 2026 [35]
H20	Claude (draft)	Ground-breaking on 1 November 2025	Detail not on cited page	Reworded to "early November 2025"
H21	Claude (draft)	SiCSem approval by "the Odisha government"	Wrong approving body	Falta SEZ Unit Approval Committee [33][34]
H22	Claude (draft)	FY25 imports US$30.3 billion, growing 23% a year	Figures not on the cited page	Withdrawn; [19] removed
H23	Claude (draft)	"Measured" efficiency gains	Simulation described as measurement	Relabelled "reported / simulated"
H24	Claude (draft)	"Carmakers can ask for tests beyond AEC-Q101"	Source does not name carmakers	Reworded [25]

Error types found: invented numbers (H1, H8, H9, H10), wrong calculation (H7), outdated information (H2, H13, H15, H19), internal contradiction (H2), conflation (H11), missing or uncheckable citations (H3, H12), incomplete answers (H5, H6), and citations that did not support the claim (H16-H18, H20-H24). No tool was error-free. Gemini, which gave no citations, had the most unsupported numbers; the AI-assisted draft itself needed 9 corrections after the final source re-check.

6. Source log summary

The full log is in *Capstone12_Source_Log_v2.xlsx*. For each URL it records ID, URL, source type, claim supported, date accessed, trust level, verification status, open action, the 24-25 October re-check result and the fix applied.

- **Used (39):** government and research sources first (NITI Aayog, PMO and PIB releases, Rajya Sabha reply, US DOE, a research paper), then company and market-research pages (TrendForce, McKinsey, NXP, TI, ADI, Toshiba), then trade and news pages. 36 are marked Opened and confirmed.
- **Not used (9):** a broker blog on the Tata fab; two exam-preparation sites and a blog on ISM project counts; a vendor blog claiming 3-5% efficiency; a 2023 article with outdated share data; a Digitimes tag page; a market-research marketing page; and the withdrawn Tele.net page [19].
- **Still flagged (3):** [5], [16] and [23] could not be re-opened in the final check; they support minor details only.

7. Prompt library

P1-P3 are the research questions as asked of each tool. P4-P11 are the working prompts for checking, drafting and design.

ID	Purpose	Prompt (template)	Improvement made
P1	RQ1 research	"Which chips are used in an EV (power, BMS, motor control), and what advantage do SiC/GaN give over silicon in efficiency and cost?"	Added "cite a URL for every figure" after Perplexity skipped the advantage part
P2	RQ2 research	"Who controls the EV chip supply chain (Infineon, STMicro, onsemi), and what is India's position (ISM, Tata Electronics, CG Power)?"	Added "cover India explicitly in a separate table"
P3	RQ3 research	"What are the main trends and risks in the EV chip market (shortage, cost, import dependency)?"	Added "give the year of every statistic"
P4	Source check	"For each claim below, find a government, company or research page that supports it. Give URL, date and verdict: Verified, Partly, Conflict, Unsupported."	Forced one verdict per claim
P5	Conflict finder	"Compare these four answers row by row. List topics where numbers or statements disagree, and say which is more likely correct and why."	Produced conflicts C1-C8
P6	Hand check	"Recompute every percentage, sum and growth rate in this table and say whether it matches the source."	Caught the 1.4× versus 1.78× error
P7	Report draft	"Write a 6-8 page report from the verified claims only. Cite with numbers. Label scenarios and projections."	"Verified claims only" stopped AI numbers entering the text
P8	Chart design	"Draw a bar chart from these rows with the finding in the title, values on bars, derived rows labelled."	Derived rows marked as derived
P9	Slide design	"Make a 6-slide executive deck: problem, chips, supply chain, India, risks, conclusion. One idea per slide."	Limited text per slide
P10	Source log	"Add a row for each URL: ID, URL, type, claim, date, trust level, status, open action."	Feeds the automated counts
P11	Citation re-check	"Open each cited page and say whether it states this exact claim, with the figure and page date. Mark SUPPORTED, PARTLY, NOT SUPPORTED or UNREACHABLE."	Found nine drafting errors (H16-H24) before submission

8. Before-vs-after analysis

Measure	Before (raw AI answers)	After (final report)
Working source URLs	Perplexity: none; Gemini: none; Claude: 27; ChatGPT: 5	39 numbered sources, each with URL and access date
Verified claim rate	Perplexity: 9 of 20 claims verified (45%); 12 of 20 verified or partly verified (60%)	Every claim in the report is Verified or Partly verified, with limits stated
Unsourced numerical claims	Gemini: more than ten (range, battery, cost ratios, $1,200, 1.4× dies and others)	None; unsupported figures were dropped
Calculation errors	One found (1.4× versus 1.78×)	Corrected; ten hand calculations documented
Citation accuracy	Final re-check: 11 of 34 citations were outdated, partly supported, unsupported or unreachable	Fixed, replaced or withdrawn; three flagged as supporting detail
ISM project count	Conflicting: 10, 12 and six	12 approved, over ₹1.64 lakh crore, from the official 15 July 2026 release
Coverage gaps	Perplexity skipped India and the SiC advantage	Both answered, with a table of Indian projects
Currency of information	Claude: construction status from August 2025; ISM 2.0 "pending"	Updated to July-September 2026, including ISM 2.0 approval and two projects shipping

Presentation of uncertainty	Scenario and projection numbers stated as facts	US$240 billion labelled a scenario; DOE 2028 values labelled projections; simulation versus vendor claims separated

Conclusion. AI tools save time on first drafts and finding leads, but accuracy came from the extra steps: asking the same question in several tools, checking each claim, recomputing numbers, separating facts from projections and, last, re-opening every citation.

9. Responsible AI, copyright, citation and limitations

- **Copyright.** Source text is paraphrased, not copied; no passages were reproduced. Charts use numbers from cited sources and show their source line.
- **Citation practice.** Each fact carries a numbered citation. Derived numbers are labelled derived. AI answers are never cited as evidence.
- **Privacy.** Only public information was used; nothing private or confidential was uploaded.
- **Over-reliance.** AI invented numbers, mixed years and gave citations that could not be checked, even in the drafting step. Human verification was applied to every figure and citation.
- **Source bias.** Vendor and trade pages favour their own products; their numbers are treated as claims and preferably replaced by government or research sources.
- **Safety.** The project is desk research only. No mains-voltage or battery experiment was carried out; any future bench work on EV batteries or high-voltage inverters needs qualified supervision.
- **Limitations.** Three sources could not be re-opened in the final check; market shares are for 2023-2024; ISM figures change with new approvals; the SiCSem investment is unresolved; EDA/IP, foundries and ADAS chips were not covered in depth.
- **Disclosure.** Perplexity, ChatGPT, Gemini and Claude were used for research; Claude also for drafting, charts and slides. The final text was checked against sources by the student.

10. Final demo plan (about 5 minutes)

1. **Problem (30 s).** Show the three research questions and why EV chips matter.
2. **Four answers (60 s).** Show the same question answered by the four tools; point out the answer with no sources.
3. **Catch the errors (90 s).** Show the hallucination log: Gemini's 1.4× dies against the hand calculation (1.78), the invented "12-18 months", the shortage-versus-oversupply conflict, and one citation that did not support its claim (H16).
4. **Automation (45 s).** Open the source log, change a status in the drop-down, and show the counts and the open-action flag update.
5. **Outputs (60 s).** Scroll through the three report charts and the 6-slide deck.
6. **Before versus after (30 s).** Show the Section 8 table and state the main lesson: verify, then write.